AI FOR GOOD HACKATHON NORWAY 2026 · AI FOR REDUCING FOOD WASTE

Bake what you'll sell

Simple, explainable bake planning that cuts bakery waste. AI suggests. The owner decides.

Team Green

Ruhit Rafian Prinon · Rizve Hasan · Jie Biam · Anne-Kristine

PROBLEM & USER UNDERSTANDING · 20%

Every afternoon, a small bakery owner guesses how much to bake tomorrow

The user: the owner of a small, independent bakery, with no time or staff for data analysis.

Their workflow today: decide tomorrow's bake before ordering and prep, from memory or a rule of thumb like "last week plus a bit".

Why it's hard: small, unpredictable daily sales per product are hard to judge by eye.

THE COST OF GUESSING

Too much: good food in the bin.

Too little: empty shelves, lost sales.

Assumption, not yet validated with a bakery owner.

PROTOTYPE & TECHNICAL EXECUTION · 20%

One tool for the bakery’s whole day

AFTERNOON

Tomorrow’s plan

How much of each product to bake, and why. Owner approves.

ALL DAY

Inventory & expiry

Batches by use-by time, with suggested discounts as they near expiry.

BEFORE CLOSING

Giveaway bags

Pack expiring items into mixed bags for a surplus app or food bank.

AT CLOSING

End of day

Enter leftovers, get safe actions in order: discount, surplus, donate, compost.

WEEKLY

Impact

Track waste, sell-outs and food rescued over time.

AI only where it helps: a demand model per product suggests the bake quantity. Expiry, discounts and food safety use simple, fixed rules.

PROTOTYPE · TOMORROW'S PLAN

A bake quantity per product, with the reason in plain words

Forecast from weekday, season, recent sales and local weather, shown as a likely range, not a single "true" number.

Bake quantity weighs the profit lost by selling out against the cost of each leftover.

Owner picks the goal: Most profit, Balanced or Least waste. Then approves, edits or resets.

WHAT THE OWNER SEES, PER PRODUCT

- Suggested bake quantity
- Expected sales and likely range
- What drove it: weekday, season, weather
- How often this covers demand
- Cost of a sell-out vs a leftover
- "Bake to order" for slow sellers

PROTOTYPE · INVENTORY, BAGS, END OF DAY

What is baked gets sold, shared or composted safely

Discount on time

Batches sorted by use-by. As selling time runs out, it suggests a discount and moving items to the front. The demo clock shows how advice shifts through the day.

Giveaway bags

Expiring items packed into mixed bags at a target value, to sell on a surplus app or give free to a food bank.

Safe by default

Leftovers get actions in order of preference. Expired items are compost only, never sold or donated.

SOCIAL & ENVIRONMENTAL IMPACT · 30%

Who benefits, what changes, and how we'll prove it

WHO BENEFITS

Bakery owners: less money spent on bread nobody buys.

Food banks and community fridges: a steady flow of safe surplus.

The climate: less food produced only to be thrown away.

WHAT CHANGES

Prevent: bake closer to real demand.

Rescue: discount, bag or donate food before it expires.

Last resort: compost only what is no longer safe.

HOW WE'LL MEASURE IT

The app already records, every day:

- units wasted
- units short (sell-outs)
- units sold off or donated

In a pilot, compare these with the owner's current routine.

The demo runs on public sample data, so we claim no impact figures until a real pilot.

RESPONSIBLE AI · 20%

AI suggests. The owner decides.

Human in control

Nothing is baked, discounted or donated automatically. Every plan needs approval.

Transparent sources

Each number shows its drivers and range. Data sources and limitations have their own page in the app.

Public data, no personal data

Public sample sales data only. Customer IDs are dropped at import, never stored.

Failure safeguards

A wrong forecast costs a few units, and the owner can edit it. Expired food is never sold or donated.

LIMITATIONS & NEXT STEP

What it doesn’t know yet, and what we’d do next

- Demo uses public sample data, not a real bakery’s sales
- Prices, costs and shelf lives are illustrative defaults
- No holidays, events or promotions in the model
- No sign-in yet: run on a trusted network only
- Open-Meteo is free for non-commercial use only

NEXT STEP

Pilot with one Oslo bakery for 4 weeks on its own sales data

Measure real waste and sell-outs against today’s routine.

Built today by Team Green with [AI tools used]. Data: Kaggle Groceries (public), Open-Meteo archive. [Disclose any pre-built components]